ECON 5021 Unit 2: SAQ Flashcards (Short)

Full question on top; the bare-minimum answer to recall is in red. Longer versions are in `ECON5021_Unit2_SAQ_Flashcards.pdf`.

1. L8 Q1

$\mathrm{E}(y) = \mu$. Show the a that minimizes $\mathrm{E}[(y-a)^2]$ is μ.

$\mathrm{E}(y-a)^2 = \mathrm{E}(y-\mu)^2 + (\mu-a)^2$, because the cross term is $2(\mu-a)\mathrm{E}(y-\mu) = 0$. Minimized at $a = \mu$.

2. L8 Q2

$m(x) = \mathrm{E}(y|x)$, $y = m(x) + u$. (a) Show $\mathrm{E}(u|x) = 0$. (b) Show $\mathrm{E}(h(x)u) = 0$. (c) Show $g(x) = m(x)$ minimizes $\mathrm{E}[(y-g(x))^2]$.

(a) $\mathrm{E}(u|x) = m - m = 0$. (b) LIE: $\mathrm{E}(hu) = \mathrm{E}[h\,\mathrm{E}(u|x)] = 0$.
(c) $\mathrm{E}(y-g)^2 = \mathrm{E}u^2 + 2\underbrace{\mathrm{E}[u(m-g)]}_{0 \text{ by (b)}} + \mathrm{E}(m-g)^2$, minimized at $g = m$.

3. L8 Q3

FOCs $-2\sum_i x_{ik}(y_i - x_i'b) = 0\ \forall k$, $e_i = y_i - x_i'b$, $x_{i1} = 1$. Show (a) $\sum e_i = 0$, (b) $\bar{y} = \bar{x}'b$, (c) $\widehat{\mathrm{Cov}}(x_k, e) = 0$.

FOC: $\sum x_{ik}e_i = 0$. (a) Take $k = 1$. (b) $0 = \bar{e} = \bar{y} - \bar{x}'b$. (c) $\widehat{\mathrm{Cov}} = \frac{1}{n}\sum x_{ik}e_i - \bar{x}_k\bar{e} = 0 - 0$.

4. L8 Q4

$X = [\iota\ \ e]$ (education), $Y = w$. (a) Show $b = \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$, showing the multiplication. (b) Show $b_2 = \widehat{\mathrm{Cov}}(e,w)/\widehat{\mathrm{Var}}(e)$.

(a) $X'X = n\begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}$ and $X'Y = n\begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$, so the n's cancel.
(b) 2×2 inverse $= \frac{1}{\overline{ee}-\bar{e}^2}\begin{bmatrix} \overline{ee} & -\bar{e} \\ -\bar{e} & 1 \end{bmatrix}$; row 2: $b_2 = \frac{\overline{ew}-\bar{e}\bar{w}}{\overline{ee}-\bar{e}^2} = \frac{\widehat{\mathrm{Cov}}}{\widehat{\mathrm{Var}}}$.

5. L9 Q1

$Y = X\beta + \boldsymbol{\varepsilon}$, $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$, $b = (X'X)^{-1}X'Y$. Show (a) $\mathrm{E}(b|X) = \beta$, (b) $\mathrm{E}(b) = \beta$.

(a) $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$, so $\mathrm{E}(b|X) = \beta + (X'X)^{-1}X'\mathrm{E}(\boldsymbol{\varepsilon}|X) = \beta$. (b) LIE: $\mathrm{E}b = \mathrm{E}[\mathrm{E}(b|X)] = \beta$.

6. L9 Q2

(a) Show $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$. (b) Show $\mathrm{Var}(b|X) = (X'X)^{-1}X'\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X]X(X'X)^{-1}$. (c) Why is $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X] = \mathrm{Var}(\boldsymbol{\varepsilon}|X)$? (d) Homoskedasticity + no autocorrelation $\Rightarrow \sigma^2 I \Rightarrow \mathrm{Var}(b|X) = \sigma^2(X'X)^{-1}$.

(a) Substitute Y. (b) $b - \beta = A\boldsymbol{\varepsilon} \Rightarrow A\,\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X)A'$. (c) $\mathrm{Var} = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') - \mathrm{E}\boldsymbol{\varepsilon}\,\mathrm{E}\boldsymbol{\varepsilon}'$, and $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$.
(d) Diagonal σ^2, off-diagonal 0, so $\sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

7. L9 Q3

$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$, $w_i = x_i\varepsilon_i$. (a) Show $b = \beta + (X'X/n)^{-1}\bar{w}$. (b) What does the LLN imply? (c) What does the CLT imply?

(a) $X'\boldsymbol{\varepsilon} = n\bar{w}$. (b) LLN: $\bar{w} \to 0$, so $b \to \beta$ (consistent).
(c) CLT: $b \overset{a}{\sim} N(\beta, \sigma^2(X'X)^{-1})$ even without normal errors.

8. L10 Q1

$H = (e_1, \ldots, e_n)'$. (a) Define P_X. (b) Show $P_X H = 0$. (c) Say (b) in words. (d) Regress e on X; show $a = 0$.

(a) $P_X = X(X'X)^{-1}X'$. (b) $P_X(Y - Xb) = Xb - Xb = 0$. (c) Residuals are orthogonal to X.
(d) $a = (X'X)^{-1}X'(Y - Xb) = b - b = 0$.

9. L10 Q2

$Y = \tau Z + \kappa X + \boldsymbol{\varepsilon}$ with Z a column of ones. Describe completely (dimensions, every element): (a) $P_Z = Z(Z'Z)^{-1}Z'$, (b) $P_Z X$, (c) X^* in $M_Z Y = \kappa M_Z X + M_Z \boldsymbol{\varepsilon}$.

(a) $n \times n$, every entry $1/n$. (b) $n \times 1$, every entry $\bar{x}$. (c) $n \times 1$, entry i is $x_i - \bar{x}$.

10. L11 Q1

$b|X \sim N(\beta, \sigma^2(X'X)^{-1})$. Explain how the standardized estimate is built and give its distribution.

$\frac{b_k - \beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1)$; swapping in $s^2 = \frac{e'e}{n-K}$ gives $\frac{b_k - \beta_k}{\widehat{SE}} \sim t_{n-K}$.

11. L11 Q2

$M_X = I - X(X'X)^{-1}X'$, $e = M_X \boldsymbol{\varepsilon}$. (a) Show $e'e = \boldsymbol{\varepsilon}' M_X \boldsymbol{\varepsilon}$. (b) Given $\mathrm{E}(e'e|X) = \mathrm{tr}(M_X \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X))$, show $\mathrm{E}\left(\frac{e'e}{n-K}\middle|X\right) = \sigma^2$.

(a) $e'e = \boldsymbol{\varepsilon}' M' M \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}' M \boldsymbol{\varepsilon}$ (symmetric: $M' = M$; idempotent: $MM = I - 2P + P = M$).
(b) $\mathrm{tr}(M\sigma^2 I) = \sigma^2[n - \mathrm{tr}((X'X)^{-1}X'X)] = \sigma^2(n-K)$; divide by $n-K$.

12. L11 Q3

$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$, $\mathrm{Var}(b|X) = \sigma^2(X'X)^{-1}$. (a) Derive $\mathrm{Var}(b_2|X)$. (b) Name the three factors.

(a) $(2,2)$ entry of $(X'X)^{-1}$: $\frac{\sigma^2 n}{n\sum x_i^2 - (\sum x_i)^2} = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$.
(b) Larger σ^2 increases it; larger n and more spread in x decrease it.

13. L11 Q4

$y_i = \beta_1 + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i$. (a) Write $\mathrm{Var}(b_3|X)$. (b) Explain each of the 3 components and when it improves precision.

(a) $\frac{\sigma^2}{(1-R_3^2)SST_3}$. (b) More precise with a smaller σ^2 (noise), a bigger SST_3 (variation in x_3, n), or a smaller R_3^2 (less correlation with the other x's).

14. L11 Q5

(a) What is multicollinearity and how does it affect precision? (b) Why can dropping a highly correlated regressor improve precision but cause bias?

(a) High R_k^2 makes $1 - R_k^2 \to 0$, so the variance and SE grow (still unbiased).
(b) Dropping the regressor lowers the variance, but it moves into ε, so $\mathrm{E}(\varepsilon|X) \neq 0$: omitted variable bias.

15. L12 Q1

$\log(fare) = 4.666 + 0.043\, dist$ ($dist$ in 100s of miles). (a) Write $fare$ in terms of $dist$. (b) Find $\partial fare/\partial dist$. (c) Show 1 mile $\Rightarrow \approx 0.043\%$ higher fare.

(a) $fare = e^{4.666 + 0.043 dist}$. (b) $0.043\, fare$. (c) $100 \frac{\partial fare}{fare} = 4.3\, \partial dist = 4.3(0.01) = 0.043\%$.

16. L12 Q2

$fare = 41 + 66 \log(dist)$. (a) Find $\partial fare/\partial dist$. (b) Show a 1% increase in $dist \Rightarrow +\$0.66$. (c) Show this does not depend on the units of $dist$ (e.g. miles).

(a) $66/dist$. (b) $\partial fare = \frac{66}{100}(100 \frac{\partial dist}{dist}) = 0.66$ per 1%.
(c) $41 + 66\log(m/100) = (41 - 66\log 100) + 66 \log m$: only the intercept changes.

17. L12 Q3

$\log(fare) = 4.25 + 0.40\log(dist)$. (a) Write $fare$ in terms of $dist$. (b) Find $\partial fare/\partial dist$. (c) Show a 10% increase in $dist$ $\Rightarrow\approx$4% higher fare.

(a) $fare = e^{4.25} dist^{0.40}$. (b) $0.40\, fare/dist$. (c) $\%\Delta fare = 0.40 \times \%\Delta dist = 0.40 \times 10 = 4\%$.